Ant Colony Optimization.py - C:/Users/BMSCECSE-L3-10/Desktop/1BM24CS037/Ant Colony Optimization.py (3.13.9)

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```
import random

network = {
    'A': {'B': 2, 'C': 4},
    'B': {'A': 2, 'C': 1, 'D': 5},
    'C': {'A': 4, 'B': 1, 'D': 2, 'E': 3},
    'D': {'B': 5, 'C': 2, 'E': 1},
    'E': {'C': 3, 'D': 1}
}

source = input("Enter source: ")
destination = input("Enter destination: ")

n_ants = 10
iterations = 50
alpha = 1
beta = 2
rho = 0.5
Q = 100

pheromone = {}
for node in network:
    for neighbor in network[node]:
        pheromone[(node, neighbor)] = 1.0

best_path = None
best_cost = float('inf')

for iteration in range(iterations):
    routes = []

    for ant in range(n_ants):
        current = source
        path = [source]
        cost = 0

        while current != destination:
            neighbors = [
                node for node in network[current]
                if node not in path
            ]

            if not neighbors:
                break

            weights = []
            for node in neighbors:
                distance = network[current][node]
                weight = (
                    pheromone[(current, node)] ** alpha
                    * (1 / distance) ** beta
                )
                weights.append(weight)

            next_node = random.choices(
                neighbors, weights=weights, k=1
            )[0]

            cost += network[current][next_node]
            path.append(next_node)
            current = next_node

        if current == destination:
            routes.append((path, cost))

            if cost < best_cost:
                best_cost = cost
                best_path = path[:]

    for edge in pheromone:
        pheromone[edge] *= (1 - rho)

    for path, cost in routes:
        deposit = Q / cost

        for i in range(len(path) - 1):
            edge = (path[i], path[i + 1])
            reverse_edge = (path[i + 1], path[i])

            pheromone[edge] += deposit
            pheromone[reverse_edge] += deposit

if best_path:
    print("Best packet route:", " -> ".join(best_path))
    print("Total routing cost:", best_cost)
else:
    print("No route found!")
```

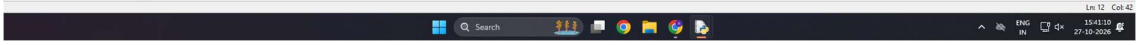
Ln: 12 Col: 42

IDLE Shell 3.13.9

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```
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/BMSCECSE-L3-10/Desktop/1BM24CS037/Ant Colony Optimization.py
Enter source: A
Enter destination: E
Best packet route: A -> B -> C -> D -> E
Total routing cost: 6
>>>
```

Ln: 9 Col: 0



Ant Colony Optimization.py - C:/Users/BMSCECSE-L3-10/Desktop/1BM24CS037/Ant Colony Optimization.py (3.13.9)

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    routes = []

    for ant in range(n_ants):
        current = source
        path = [source]
        cost = 0

        while current != destination:
            neighbors = [
                node for node in network[current]
                if node not in path
            ]

            if not neighbors:
                break

            weights = []
            for node in neighbors:
                distance = network[current][node]
                weight = (
                    pheromone[(current, node)] ** alpha
                    * (1 / distance) ** beta
                )
                weights.append(weight)

            next_node = random.choices(
                neighbors, weights=weights, k=1
            )[0]

            cost += network[current][next_node]
            path.append(next_node)
            current = next_node

        if current == destination:
            routes.append((path, cost))

            if cost < best_cost:
                best_cost = cost
                best_path = path[:]

    for edge in pheromone:
        pheromone[edge] *= (1 - rho)

    for path, cost in routes:
        deposit = Q / cost

        for i in range(len(path) - 1):
            edge = (path[i], path[i + 1])
            reverse_edge = (path[i + 1], path[i])

            pheromone[edge] += deposit
            pheromone[reverse_edge] += deposit

if best_path:
    print("Best packet route:", " -> ".join(best_path))
    print("Total routing cost:", best_cost)
else:
    print("No route found!")
```

Ln: 12 Col: 42

IDLE Shell 3.13.9

File Edit Shell Debug Options Window Help

```
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (
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>>>
= RESTART: C:/Users/BMSCECSE-L3-10/Desktop/1BM24CS037/Ant Colony Optimization.py
Enter source: A
Enter destination: E
Best packet route: A -> B -> C -> D -> E
Total routing cost: 6
>>>
```

Ln: 9 Col: 0

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